Papa Amadou BA

GEN AI & AGENTIC AI CONSULTANT · DATA SCIENCE LEAD

Chelles 77500, France · +33 (0)6 20 33 22 07 · papamadouba@gmail.com · French National · English B2

● Available for Projects

AGENTIC AI STACK

Frameworks

LangChain · LangGraph · AutoGen · CrewAI

LLMs

GPT-4o · Claude 3.5 · Gemini 1.5 · LLaMA 3 · Mistral · Falcon

ML / DL

PyTorch · TensorFlow · HuggingFace · Scikit-learn · XGBoost · LoRA / QLoRA

Vector Stores / Graphs

Pinecone · ChromaDB · FAISS · Neo4j · Weaviate

Cloud / MLOps

AWS Bedrock · SageMaker · GCP Vertex AI · Azure OpenAI · Docker · MLflow · LangSmith

Programming & Data

Python · FastAPI · PySpark · PostgreSQL · MongoDB · ETL Pipelines

EDUCATION

MSc Data Science & Gen AI
Univ. of Technology, Paris · 2024
ML · DL · NLP · Generative Models

PGI/ERP – IT App Developer
DORANCO, Paris · 2021

Certificate – Data Control
AFPA, Paris · 2020

BSc Finance & Management
IFAM, Paris · 2003

CERTIFICATIONS

LangChain & LangGraph – Agentic AI
DeepLearning.AI · 2025

Multi-Agent Systems – AutoGen
Microsoft / DeepLearning.AI · 2025

AWS ML Specialty
Amazon Web Services · 2024

GCP Professional ML Engineer
Google Cloud · 2024

Generative AI with LLMs
DeepLearning.AI / AWS · 2024

LANGUAGES

French — Native

English — B1–B2

Wolof / Pulaar — Fluent

COMPETENCIES

- AI Strategy & Roadmapping
- Stakeholder Communication
- Executive Presentations
- Cross-Cultural Negotiation
- Team Leadership

PROFESSIONAL SUMMARY

Visionary **Gen AI & Agentic AI Consultant** with an MSc in Data Science & Generative AI (Paris, 2024) and 20+ years of entrepreneurial leadership. Expert in **multi-agent autonomous systems** (LangChain, LangGraph, AutoGen, CrewAI), **RAG pipelines**, and enterprise **LLM fine-tuning** (LoRA/QLoRA). Bridges cutting-edge AI research with tangible business outcomes across EU & international markets.

PROFESSIONAL EXPERIENCE

Senior Gen AI & Agentic AI Consultant — 2024 – Present

Freelance / Independent Consulting · Remote (EU & International)

- Architected **multi-agent systems** (LangGraph, AutoGen, CrewAI) for autonomous decision pipelines — cutting analyst turnaround from 3 days to 2 hours.
- Built **hybrid RAG solutions** (Pinecone, ChromaDB, FAISS) with GPT-4o & Mistral, improving answer faithfulness +38% over vanilla RAG.
- Fine-tuned **Mistral-7B via QLoRA** on French financial corpus; deployed on AWS SageMaker at 90% lower cost vs GPT-3.5.
- Delivered AI strategy workshops & POC roadmaps to C-suite on AWS Bedrock, GCP Vertex AI, and Azure OpenAI.

LangChain · LangGraph · AutoGen · CrewAI · RAG · QLoRA · AWS Bedrock · GCP Vertex AI

Data Scientist & ML Engineer — 2022 – 2024

AI Research Projects · University of Technology, Paris

- Built NLP models (BERT, T5, GPT-2) for classification & Q&A; integrated **Neo4j knowledge graph** pipelines for multi-hop reasoning.
- Deployed computer vision (CNNs, ViT) and time-series forecasting (LSTM, Prophet); managed MLOps with MLflow & DVC.

PyTorch · HuggingFace · Neo4j · MLflow · NLP

IT Developer & Data Engineer — 2020 – 2022

DORANCO & AFPA Ecosystem · Paris

- Built ETL pipelines (Python, PySpark) and RESTful microservices (FastAPI, Flask) containerised with Docker & Kubernetes.

Entrepreneur & Managing Director — 2003 – 2020

SASU CMMS – Import/Export Trade · Paris · Germany · Senegal

- Founded & scaled cross-border automotive parts company across 3 countries for 17 years; applied data-driven inventory optimisation and demand forecasting.

KEY AI PROJECTS

Multi-Agent Supply Chain Intelligence — LangGraph · CrewAI · GPT-4o · Pinecone
LangGraph state-machine with specialised agents (retriever, risk-analyser, writer) delivering end-to-end supply chain risk reports.
↓ Turnaround: 3 days → 2 hrs

Enterprise RAG Chatbot with Neo4j Knowledge Graph — LangChain · LLaMA 3 · ChromaDB
Hybrid dense-vector + graph-traversal RAG for complex multi-hop enterprise Q&A. ↑ Faithfulness +38% vs baseline

AutoGen Autonomous Research Agent — AutoGen · OpenAI API · Bing Search
Planner–Researcher–Critic–Executor team autonomously completing intelligence research tasks end-to-end. ↓ Research time –70%